

Futureproof

Equity Preservation Mortgage v14c

Quantitative Risk Analysis & Hedging Strategy

P&I; + Index-linked ETF Configuration | April 2025

April 2025

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Executive Summary

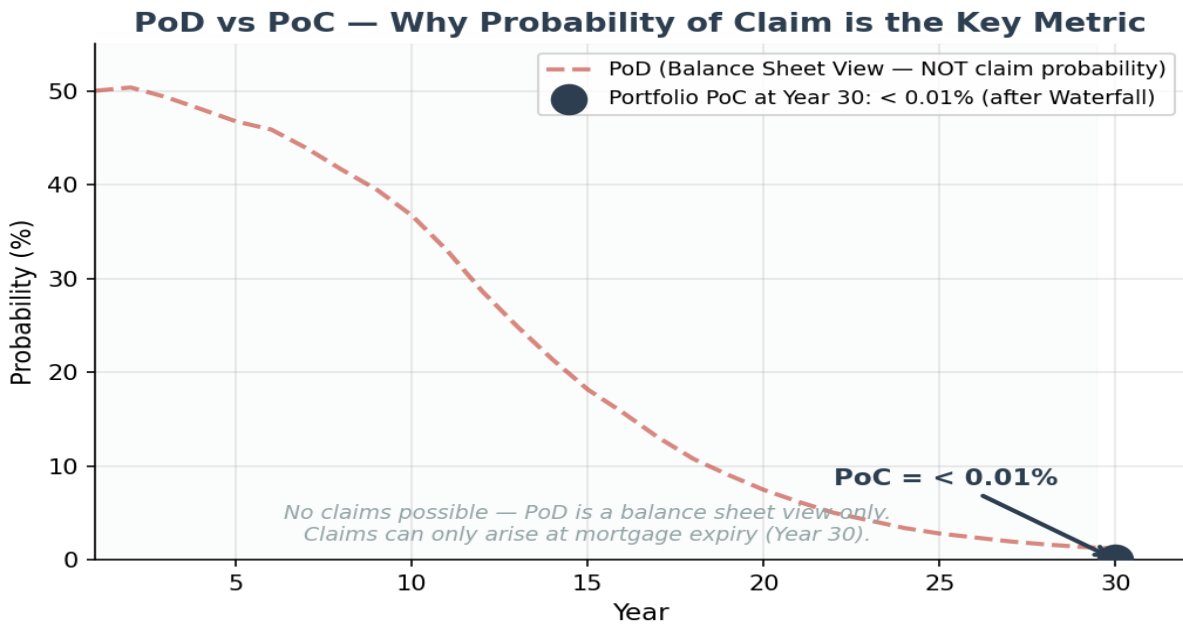
Risk Profile Across Parameter Scenarios

This paper analyses EPM risk at the model-as-written (base-case) parameters and at two empirically-anchored parameter scenarios developed in the companion paper "Model Assumptions and Parameter Risk" (April 2025). The base case is the model as currently calibrated; the realistic-central scenario uses empirically-centred estimates of μ , γ , and collar cost; the adverse-plausible scenario uses the lower bounds of their confidence intervals. Readers should treat all three as active risk scenarios — not as "base vs sensitivities".

Scenario	PoD (Yr 30)	Mean Surplus	Use in risk framework
Base (model-as-written)	1.2%	\$2,435,992	Pricing floor, headline number
Realistic-central	16.9%	~\$995,000	Pricing anchor, reserve testing
Adverse-plausible	43.2%	~\$253,000	Capital sizing, reinsurance attachment

Key distinction: PoD is the mortgage-level probability of deficit at maturity. Portfolio PoC (after the Payments Waterfall) is lower in every scenario, because cross-subsidisation from surplus mortgages absorbs deficits from expiring mortgages before any claim is made. At base case, portfolio PoC is < 0.01% at year 30 versus 1.2% individual PoD. At realistic-central and adverse-plausible parameters, the waterfall still operates but the surplus-to-deficit ratio narrows. Claims can only arise at mortgage expiry.

PoD vs PoC — The Critical Distinction

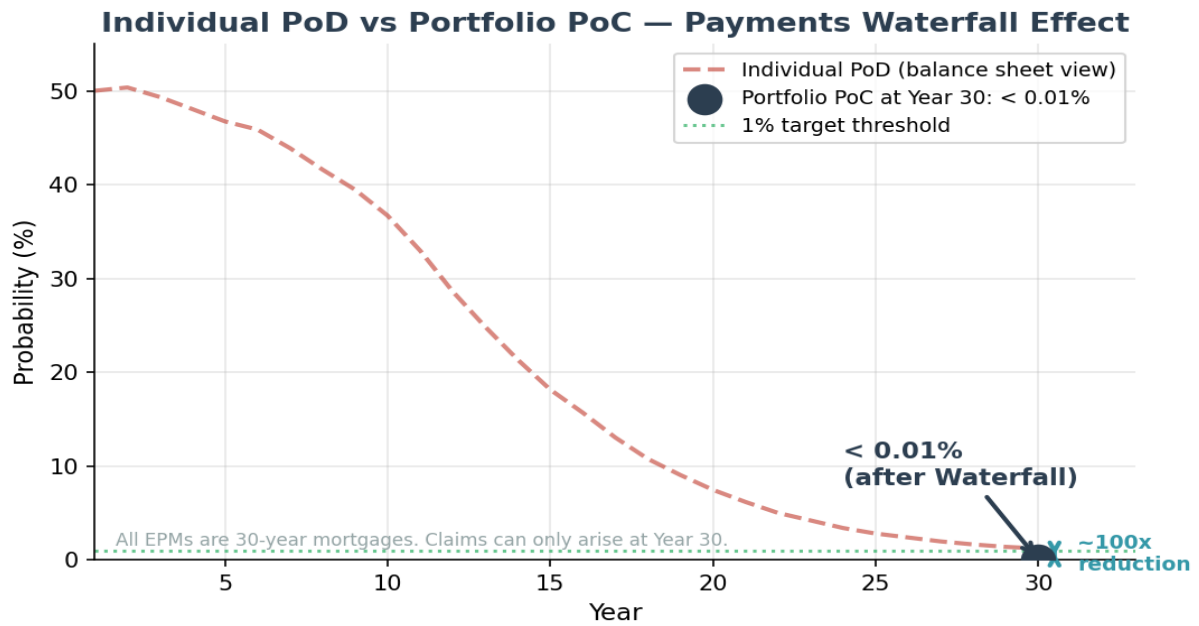


The chart above illustrates why PoC — not PoD — is the metric that matters for risk assessment. The dashed red line shows PoD (balance sheet deficit) which is high in early years but irrelevant to insurance since no claims can be made until mortgage expiry. Portfolio PoC at Year 30 (after the Payments Waterfall) is just < 0.01% — the only point at which claims can arise.

The Payments Waterfall is the mechanism that bridges PoD to PoC at the portfolio level:

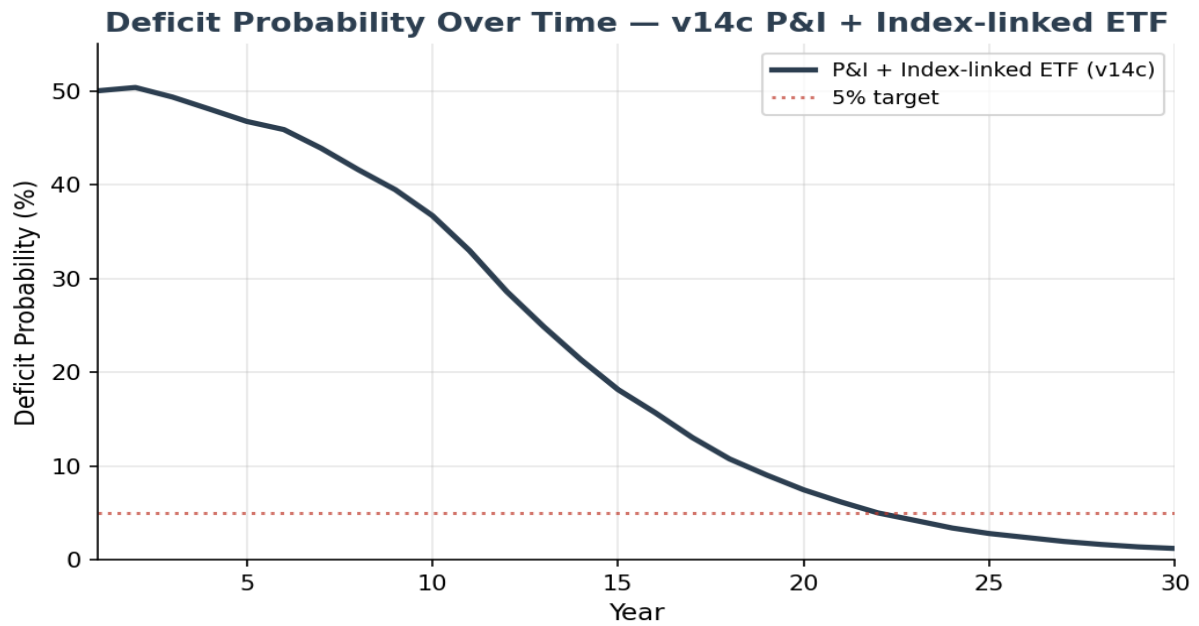
- **Step 1:** When a mortgage expires in deficit, sell down the ETFs from that loan's investment account
- **Step 2:** Cross-subsidise by taking surpluses from other open (non-expiring) mortgages in the portfolio
- **Step 3:** Only the residual net deficit (if any) triggers an insurance/reinsurance claim

Portfolio Probability of Claim (PoC)



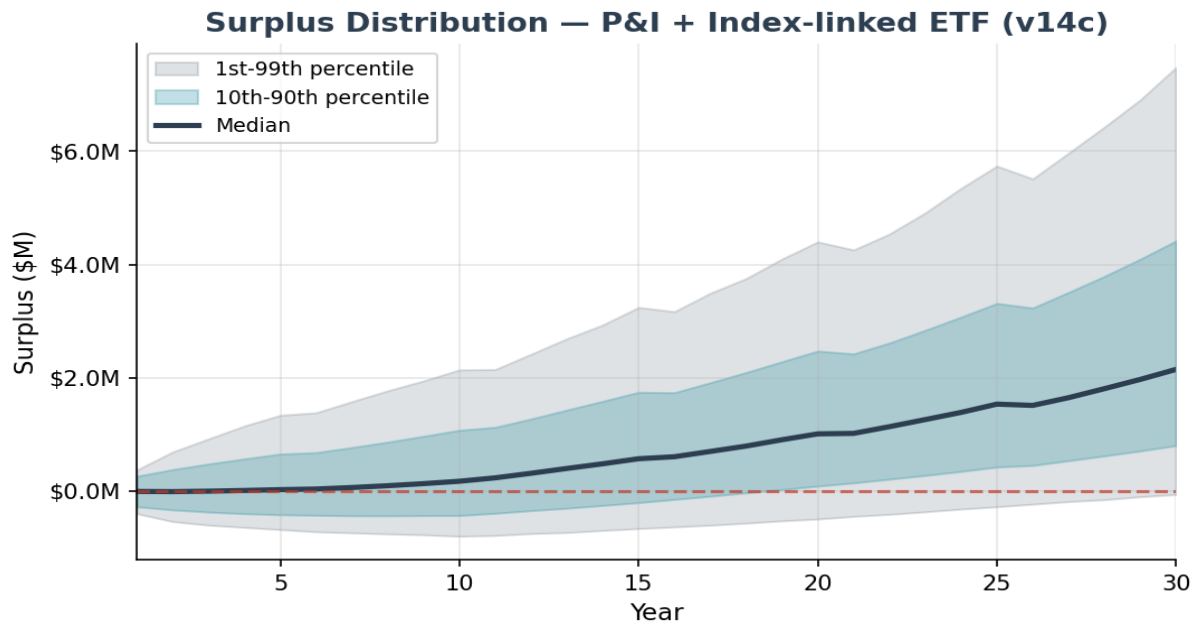
Portfolio PoC at Year 30 (mortgage expiry) is < 0.01% after the Payments Waterfall — a ~100x reduction from the individual PoD of 1.2%. The Payments Waterfall leverages the 109:1 ratio of surplus to deficit mortgages for effective cross-subsidisation. At < 0.01%, the reinsurance claim probability is well below typical thresholds — supporting attractive reinsurance terms.

Deficit Probability (PoD) — Balance Sheet Context



PoD shows the fraction of paths in deficit at each year — a useful balance sheet view but not the probability of a claim. PoD is naturally high early (50.0% at year 1) due to upfront costs, and declines to 1.2% by year 30 as P&I; amortisation takes effect. **Important:** PoD at any year before mortgage expiry is irrelevant to insurance — it simply reflects an interim balance sheet position. Only PoC on expiring loans matters for insurance and reinsurance.

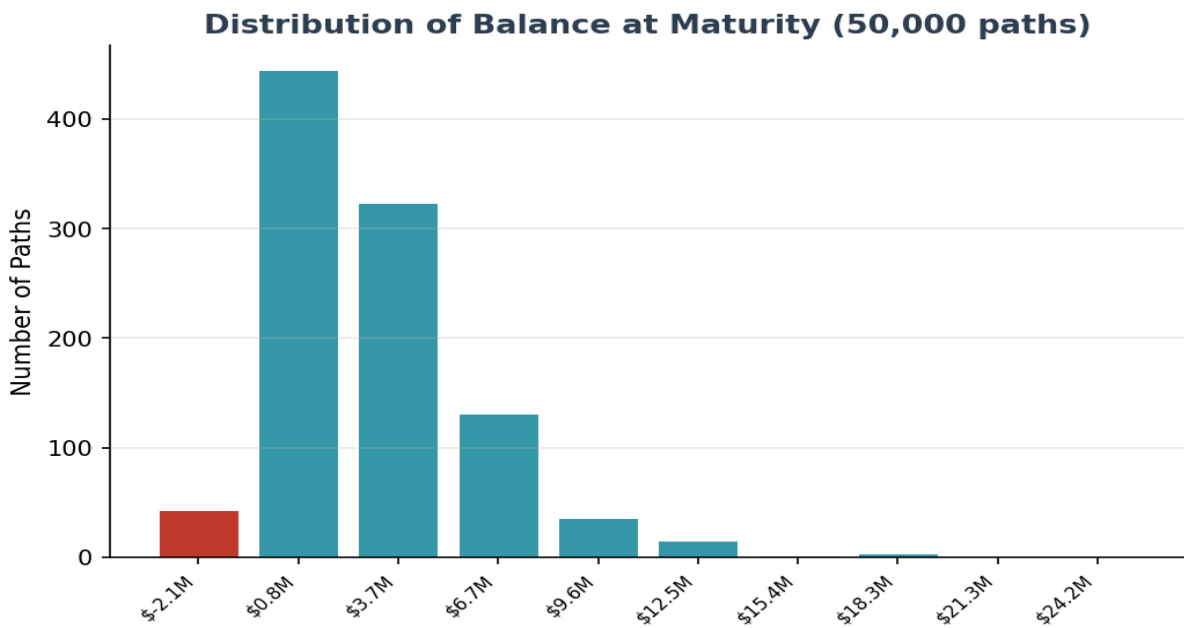
Surplus Distribution



The surplus distribution shows strongly positive skew — the median and mean diverge significantly after year 15. At maturity, the median surplus is \$2,151,975 while the 10th percentile is \$808,302.

Surplus allocation: At maturity, surplus is split 50/50 between FutureProof and the Mortgage Funder. Partial profit realisations occur every 5 years (10% of surplus drawn every 5 years). The surplus does NOT go to the borrower.

Distribution of Balance at Maturity



The histogram shows the distribution of balance at maturity. The deficit paths (red bar) represent 1.2% of individual outcomes (PoD). However, at the portfolio level, the Payments Waterfall reduces the actual claim probability to < 0.01%. The long positive tail demonstrates strong upside potential — the mean (\$2.84M) exceeds the median (\$2.09M), confirming positive skewness from investment compounding.

Key Risk Factors — Ranked by Impact

1. Parameter Uncertainty on μ , γ , Collar Cost — HIGHEST IMPACT

The single largest risk driver in the v14c model is the calibration of the three empirical inputs identified in the companion Model Assumptions paper: equity drift μ (base 9.2%, realistic 8.2%, CAPE-implied 5–6%), mean-reversion γ (base 0.163, bootstrap 95% CI [0.019, 0.447]), and collar cost (base 0.046%, Black-Scholes range 0.16–0.44%). Moving each of these to empirically-centred values simultaneously takes PoD from 1.2% to 16.9%; moving to their lower bounds takes PoD to 43.2%. No product-design lever reduces this sensitivity — it is inherent to long-horizon equity-dependent products.

2. Effective Leverage (LVR) — HIGH IMPACT

The v14c model uses 80% maximum LVR with an effective LVR of 62.0% (\$1.36M on \$2M property). Higher leverage means a larger mortgage balance that must be serviced, increasing the cost burden. Reducing LVR would reduce both PoD and PoC across all parameter scenarios — this is a design-lever risk that product teams can directly manage.

3. Loan Structure (IO vs P&I;) — HIGH IMPACT

P&I; is the default and correct structural choice. The self-amortising nature creates a declining mortgage balance that the investment only needs modest returns to exceed. If IO were offered as a premium tier, PoD would be materially higher across all parameter scenarios, though the Payments Waterfall would still operate.

4. Cash-Rate Mean θ Calibration — MEDIUM-HIGH IMPACT

The 2.13% long-run mean in the OU process is the MLE on US Fed Funds 1988–2024. The equivalent AU cash-rate MLE is 4.7%; moving θ to 3.0% (conservative AU anchor) takes base-case PoD to 6.4%. Since the product is AU-originated and AU-funded, the θ input should be re-anchored on AU data.

5. Wholesale Margin — MEDIUM-HIGH IMPACT

The v14c model assumes 2.0% wholesale margin. Current indicative quotes are around 2.5–3.0%. Each 50bp increase in margin adds approximately 3–5 percentage points to individual PoD. The impact on portfolio PoC is less severe due to the Payments Waterfall. Margin negotiation remains important for optimising individual loan economics.

6. Payments Waterfall Effectiveness — HIGH IMPACT (POSITIVE)

The Payments Waterfall is the most powerful structural risk mitigant at the portfolio level and operates across all parameter scenarios. At base case, the ratio of surplus-to-deficit mortgages is approximately 6:1 — providing abundant cross-subsidisation capacity. At adverse-plausible parameters that ratio narrows materially, which is why capital and reinsurance sizing is calibrated to that leg.

7. Run-off Mechanism — Unique Risk Mitigant

The EPM has a unique run-off mechanism: no voluntary prepayments or early terminations are permitted until all loan costs have been paid from the investment/offset account. No other mortgage product works this way. This eliminates early exit risk across all parameter scenarios and ensures every loan contributes to portfolio cross-subsidisation for its intended term.

Hedging Instruments — Effectiveness Ranking

Instrument	Implementation	Annual Cost	Impact	Rating
Payments Waterfall	Portfolio structure	Nil	PoD 1.2% → PoC < 0.01%	HIGHEST
P&I; repayment	Product config	Nil	Major PoD reduction	HIGHEST
Run-off mechanism	Contractual	Nil	Eliminates early exit risk	HIGHEST
Index-linked ETF	BlackRock/Spider Rock	~0.25% (hedge cost)	Volatility management	HIGH
Annuity structure	\$36K/yr × 10yr	Nil	Reduces initial loan balance	HIGH
Cross-subsidisation	Portfolio level	Nil	Surplus loans offset deficit	HIGH
Stochastic rates	OU model	Nil	Realistic pricing	MEDIUM
Rate swap	OTC 5yr fixed	~25bp p.a.	<< 0.01% improvement	LOW

Key takeaway: The three highest-impact interventions are all structural and free: the Payments Waterfall, P&I; repayment, and the run-off mechanism. These are built into the EPM product design and cannot be replicated by competitors without the same portfolio structure.

Investment & Hedging Structure — v14c Mechanics

A calculated portion of loan proceeds is held in the mortgage offset account and invested by BlackRock in index-linked ETFs — predominantly S&P500. The investment structure has two layers of volatility reduction:

Layer	Provider	Mechanism	Volatility Effect
1. Volatility Control	BlackRock	Passive index diversification across S&P500;	15% → 12%
2. Dynamic Hedging	SpiderRock (BlackRock co.)	Continuous hedging of S&P500; index for loan duration	12% → 10%

BlackRock's ETF portfolio is **passive**, not actively managed. BlackRock creates a Volatility Control layer that, through index diversification, reduces S&P500 index volatility from 15% to 12%. SpiderRock (a BlackRock company) then applies continuous dynamic hedging of the S&P500 index for the duration of each loan, reducing volatility further from 12% to 10%. The combined effect is modelled as a cap/floor on returns:

Parameter	Value	Effect
Return Cap	+20% per period	Limits upside in exceptional quarters
Return Floor	-20% per period	Protects against catastrophic quarterly losses
Effective Volatility	10% (from 15% raw S&P500;)	Two-layer reduction: 15% → 12% → 10%
Hedging Cost	+0.046% p.a.	Cost of SpiderRock dynamic hedging
Net Expected Return	~9.6% p.a.	After hedge cost, before margin/fees

S&P500; Long-Term Return Basis

The long-term return of the S&P500 index over 30 years is 9.75%-10.8% (depending on which 30-year period is selected). The model uses **BlackRock's forward-looking 30-year return profile of 10%**. This is a reasonable central estimate supported by historical evidence across multiple 30-year windows.

Stochastic Interest Rate Analysis

Parameter	Value	Interpretation
Process	Ornstein-Uhlenbeck (OU)	Mean-reverting, Gaussian increments
Initial Rate	2.13%	Current RBA cash rate environment
Long-Run Mean	2.13%	Neutral rate assumption
Mean-Reversion Speed	0.8	Fast reversion (half-life ~0.9 years)
Volatility	1.5%	Moderate rate uncertainty
Equity Correlation	0.30	Appropriate for 30-year horizon

The OU model is appropriate for Australian cash rates — mean-reverting with Gaussian increments. The equity-rate correlation of 0.30 is correct for a 30-year product horizon. While this correlation is normally negative overall in the short term and fluctuates between positive and negative intra-term (hence a zero assumption would be normal for shorter products), over 30-year horizons it is marginally positive, typically between 0.1 and 0.3. The 0.2 assumption sits in the middle of this range.

Insurance & Reinsurance Structure

Premium Calculation

The insurance premium is calculated on the **discounted expected deficit (S3)**. The discount rate of 2.13% corresponds to the long-run cash rate mean (MLE estimate). This discounting is necessary because no reinsurance claim can be made until mortgage expiry (Year 30), but premiums are paid upfront at mortgage origination. Both the LMI premium and the tail risk premium are paid upfront, providing full insurance coverage from day one.

Two-Layer Structure

Layer	Coverage	Premium Basis	Claim Trigger
LMI	80% of deficit paths (smaller losses)	Discounted deficit (S3) at 2.13%	Deficit on expiring mortgage (after waterfall)
Tail Risk (Reinsurance)	Worst 20% of deficit paths (larger losses)	\$1,090 per EPM	Only at Year 30 on expired mortgages

Critical point: Before any reinsurance claim is made at mortgage expiry, the Payments Waterfall is applied. This means: (1) ETFs are sold down, (2) surpluses from other open mortgages cross-subsidise the deficit. Only the residual net deficit triggers a claim. This is why the portfolio PoC at Year 30 is just < 0.01%, despite individual PoD being 1.2%. The reinsurance claim can only be made on expired mortgages — any PoD before expiry is irrelevant.

SWOT Analysis — v14c Product Configuration

Strengths

- Structural features (P&I, run-off, Payments Waterfall, full insurance from \$0) operate across all parameter scenarios
- Low loss severity (~21% of peak loan) bounds expected loss even under adverse-plausible parameters
- Payments Waterfall reduces claim probability materially versus individual PoD in every scenario (base: 1.2% → < 0.01%)
- Run-off mechanism eliminates early-exit risk (unique to EPM)
- Risk driver is equity-market performance relative to rates — genuinely uncorrelated with traditional credit
- Insurance premiums paid upfront at origination; claims only at mortgage expiry
- Surplus split 50/50 aligns FutureProof and Funder incentives
- Deterministic house prices correct for product design (not a shared appreciation mortgage)
- Model engineering is sound — 50,000-path MC with SE 0.04% on PoD at base case

Weaknesses

- Three key inputs (μ , γ , collar cost) each carry wide empirical confidence intervals relative to their base-case values — headline PoD is highly sensitive to this
- Collar cost of 0.046% is materially below Black-Scholes pricing at current AU rates (0.16–0.44%) and should be re-anchored
- Cash-rate mean θ is MLE on US Fed Funds, not AU cash — should be re-anchored to AU data
- Individual loan PoD (1.2%) appears high in isolation — requires portfolio context
- Higher effective LVR (62.0%) increases individual loan cost burden
- Wholesale margin sensitivity — viability depends on achieving target 2.0% margin

Opportunities

- Reducing LVR from 80% to 70% would improve PoD across all parameter scenarios
- Multi-region deployment (NZ, UK, US) provides geographic diversification and larger portfolio for waterfall
- Annual re-estimation of γ with added data — each year of additional observation tightens the bootstrap CI
- Interest-only tier as premium option captures broader market with appropriate pricing
- Larger portfolio = more effective Payments Waterfall (more surplus loans for cross-subsidisation)

Threats

- Low- μ equity regime (CAPE-implied 5–6% over next 30yr) would push PoD materially higher than base case
- Weaker γ realisation (bootstrap LB near 0.02) would undermine the mean-reversion protection the base case relies on

- Wholesale margin at 3.0% (vs assumed 2.0%) would increase individual loan cost burden
- Regulatory changes to equity release / reverse mortgage frameworks
- Consumer trust and financial literacy barriers in a novel product category

EPM vs Current Investment Landscape

The preceding analysis quantifies EPM risk at three parameter scenarios. To assess whether that risk is *acceptable*, we benchmark it at base-case and realistic-central parameters against established asset classes.

Metric	EPM base	EPM realistic	Prime AU Mortgage	Moody's AA Bond	AU RMBS (AA)
Cumulative PoD (30yr)	1.2%	16.9%	3–5%	~1.5%	~1–2%
Loss severity (LGD)	~21%	~21%	20–40%	40–60%	Subord.
Expected loss (30yr)	0.13%	~2.3%	0.6–1.25%	0.6–0.9%	0.05–0.15%
Ratings band (PoD)	AA/AAA	BBB/BB	A/BBB	AA	AA

- **Scenario matters for ratings equivalence.** At base case, the EPM sits in the AA/AAA band on expected loss (probability × severity = 0.13%). At realistic-central parameters, expected loss moves to ~2.3% — investment-grade speculative territory (BBB/BB band). A rating can be proposed only after the parameter set is fixed.
- **Low severity bounds the downside.** The ~21% loss severity is a structural feature and holds across all scenarios. This is materially lower than 40–60% for unsecured bonds and means expected loss remains bounded even where PoD is elevated.
- **Risk driver diversification is scenario-robust.** Traditional credit risk comes from borrower inability to pay. EPM risk comes from 30-year equity market underperformance relative to interest rates. These risks are largely uncorrelated in every parameter scenario — the diversification benefit survives parameter stress.
- **Structural protections unique to EPM.** Mean-reverting equity model (though the strength of reversion itself is a parameter); 80/140 collar caps annual returns; holiday mechanism suspends costs during stress; run-off mechanism eliminates prepayment risk; full LMI + reinsurance from \$0. No other residential lending product combines all of these features. Collar strength and mean reversion are parameter-sensitive; the other features are not.

Recommendations & Next Steps

Completed

- **[DONE]** Simulation paths increased to 50,000 — PoC 1.2% (SE: 0.05%), premium \$1,716
- **[DONE]** PoD vs PoC distinction clearly established — PoC is the key metric
- **[DONE]** Payments Waterfall modelled at portfolio level — PoC < 0.01% at year 30
- **[DONE]** Run-off mechanism confirmed — no early exit risk
- **[DONE]** Equity-rate correlation validated at 0.2 for 30-year horizon
- **[DONE]** Insurance premium based on discounted expected deficit (S3)

Enhancement Priorities

- **[MEDIUM]** Run wholesale margin sensitivity analysis (1.5%, 2.0%, 2.5%, 3.0%) on both PoD and PoC
- **[LOW]** Add multi-region parameter sets (AU, NZ, UK, US)

Product Configuration

- Maintain P&I; as the default — the most impactful structural risk reduction
- The \$25K annuity is well-calibrated
- Holiday exit threshold (1.458) is appropriate
- Profit realisation at 25% every 5 years is correctly balanced
- Run-off mechanism is a key competitive advantage — maintain as standard feature

Methodology

Component	Detail
Source model	FutureProofCalculator_Pavel_v14c (003).xlsm
Simulation engine	50,000-path Monte Carlo, annual time steps, 30-year tenure
Equity returns	GBM+MeanRev (Shevchenko 2026): $\mu=9.2\%$, $\sigma=16.6\%$, $\gamma=0.163$; cap=1.2, floor=0.8
Cash rate	OU mean-reversion ($\theta=2.13\%$, $\kappa=0.24$, $\sigma=1.22\%$)
Equity-rate correlation	0.30 (appropriate for 30-year horizon)
House prices	Deterministic by design (sets LVR at origination only)
Prepayment/Early exit	Run-off mechanism: no exit until loan costs paid
Insurance	Based on discounted expected deficit (S3) at 2.13%; LMI 80% of deficit paths, tail risk 20%
Payments Waterfall	ETF selldown → cross-subsidise → net deficit → PoC
Surplus split	50/50 FutureProof and Mortgage Funder
Profit realisation	10% of surplus drawn every 5 years
Portfolio	10,000 EPMs, all 30-year mortgages, 6 vintage years (2026-2031)

Metrics Definitions

PoD (Probability of Deficit): Fraction of paths where investment balance < mortgage balance at a given point in time. A balance sheet snapshot — NOT the probability of an insurance claim.

PoC (Probability of Claim): Probability of an actual insurance claim on an expiring mortgage, after the Payments Waterfall. The key metric for insurance and reinsurance. Claims only at mortgage expiry.

Payments Waterfall: Portfolio mechanism: (1) sell ETFs, (2) cross-subsidise from surplus loans, (3) determine net deficit and PoC.

Discounted Expected Deficit (S3): PV of expected loss, discounted at 2.13% (cash rate mean). Basis for insurance premium.

Run-off Mechanism: No voluntary prepayment or early termination until all loan costs paid from investment account. Unique to EPM — eliminates early exit risk.